

Curriculum Vitae — Nico Lang

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Research experience

03/2025– **Assistant Professor** — Computer Vision and Machine Learning for Earth Observation, Department of Computer Science (DIKU), University of Copenhagen; Global Wetland Center; Pioneer Centre for AI
09/2022–03/2025 **Postdoctoral Researcher**, DIKU, University of Copenhagen; Pioneer Centre for AI
02/2018–05/2022 **PhD Research Assistant**, Photogrammetry and Remote Sensing, ETH Zürich
03/2013–10/2017 **Student Researcher**, Photogrammetry and Remote Sensing, ETH Zürich
02/2015–04/2015 **Intern**, LOGITECH UPICTO GmbH
10/2014–01/2015 **Intern**, upicto GmbH (ETH spin-off)

Academic degrees

05/2022 Doctor of Sciences, ETH Zürich
09/2017 MSc in Geomatics, ETH Zürich
08/2014 BSc in Geomatik und Planung, ETH Zürich

Scientific focus areas

Computer vision; Deep learning; Machine learning; Remote sensing; Earth observation; Environmental monitoring; Global vegetation structure mapping; Representation learning; Self-supervised learning, Open-world recognition; Uncertainty quantification; Fine-grained recognition

Selected professional activities

2026– **ELLIS member**
2025– **Area Chair** for CVPR 2026 and ECCV 2026
2025– **Program Committee** (Keynotes) *Nordic AI Meet 2025*
2025– **Co-director** *P1 program on AI for Climate & Conservation*
2025– **Coordinator** *Climate & Conservation at the Pioneer Centre for AI*
2025– **Core team member** *Climate AI Nordics network*
2024 **Organizer** *PhD summer course SSL4EO-2024: Self-Supervised Learning for Earth Observation, University of Copenhagen*
2023 Drummer in the *CVPR 2023 house band*, Vancouver
Workshop organization
2025– **Organizer** *AICC: Workshop on AI for Climate and Conservation, EurIPS*
2025– **Organizer** *Advances in Representation Learning for Earth Observation (REO), EurIPS*
2024 **Organizer** *Visipedia workshop 2024*, Copenhagen
2023– **Organizer** (main point of contact) *FGVC10, FGVC11, FGVC12, FGVC13, CVPR*

Academic services

2019– **Reviewer** *CVPR [Outstanding reviewer 2023], ECCV, ICCV, TPAMI, RSE, ISPRS, Nature Comms. Env., Global Change Biology, NLDL, CVPR FGVC workshop, NeurIPS CompSust-2023 workshop, ICCV Joint workshop on marine vision, AI in Science Summit 2025, EurIPS AICC workshop, EurIPS REO workshop*
2026 **External examiner** PhD thesis by Miguel Espinosa, University of Edinburgh
2026 **External examiner** PhD thesis by Yuchang Jiang, University of Zurich
2025 **Reviewer** PhD proposal by Claire Robin, Wageningen University & Research
2025 **Reviewer** for the ELLIS PhD Program

Bibliography overview

Author of 20 peer-reviewed journal and conference articles. See [Google Scholar](#) or the separate publication list. Note that CVPR, ECCV, NeurIPS, ICLR are top-tier conferences ranked A* according to [CORE](#).

Ten selected publications

- [1] Sander Riisøen Jyhne, Christian Igel, Morten Goodwin, Per-Arne Andersen, Serge Belongie, and **Lang, Nico**. “[SuperF: Neural Implicit Fields for Multi-Image Super-Resolution](#)”. In: *ICLR*. 2026.
- [2] Yuyan Chen, **Lang, Nico**, B. Christian Schmidt, Aditya Jain, Yves Basset, Sara Beery, Maxim Larrière, and David Rolnick. “[Open-Insect: Benchmarking Open-Set Recognition of Novel Species in Biodiversity Monitoring](#)”. In: *NeurIPS Datasets and Benchmarks Track*. [Spotlight]. 2025.
- [3] Vésteinn Snæbjarnarson, Kevin Du, Niklas Stoehr, Serge Belongie, Ryan Cotterell, **Lang, Nico**, and Stella Frank. “[Taxonomy-Aware Evaluation of Vision-Language Models](#)”. In: *CVPR*. 2025.
- [4] Vishal Nedungadi, Ankit Kariryaa, Stefan Oehmcke, Serge Belongie, Christian Igel, and **Lang, Nico**. “[MMEarth: Exploring multi-modal pretext tasks for geospatial representation learning](#)”. In: *ECCV*. 2024.
- [5] **Nico Lang**, Vésteinn Snæbjarnarson, Elijah Cole, Oisín Mac Aodha, Christian Igel, and Serge Belongie. “[From coarse to fine-grained open-set recognition](#)”. In: *CVPR*. 2024.
- [6] Bingchen Zhao, **Lang, Nico**, Serge Belongie, and Oisín Mac Aodha. “[Labeled Data Selection for Category Discovery](#)”. In: *ECCV*. 2024.
- [7] Nikolai Kalischek, **Lang, Nico**, Cécile Renier, Rodrigo Caye Daudt, Thomas Addoah, William Thompson, Wilma J Blaser-Hart, Rachael Garrett, Konrad Schindler, and Jan D Wegner. “[Cocoa plantations are associated with deforestation in Côte d’Ivoire and Ghana](#)”. In: *Nature Food* (2023).
- [8] Siyu Liu, Martin Brandt, Thomas Nord-Larsen, Jerome Chave, Florian Reiner, **Lang, Nico**, Xiaoye Tong, Philippe Ciais, Christian Igel, Adrian Pascual, et al. “[The overlooked contribution of trees outside forests to tree cover and woody biomass across Europe](#)”. In: *Science Advances* (2023).
- [9] **Nico Lang**, Walter Jetz, Konrad Schindler, and Jan Dirk Wegner. “[A high-resolution canopy height model of the Earth](#)”. In: *Nature Ecology & Evolution* (2023).
- [10] **Nico Lang**, Nikolai Kalischek, John Armston, Konrad Schindler, Ralph Dubayah, and Jan Dirk Wegner. “[Global canopy height regression and uncertainty estimation from GEDI LIDAR waveforms with deep ensembles](#)”. In: *Remote sensing of environment* (2022).

Talks

	Multimodal Test-Time Adaptation
2026	Invited talk <i>CVPR 2026 workshop on Monitoring the World Through an Imperfect Lens</i> , Denver
	Learning Representations for Global Ecosystem Modelling
2026	Invited talk <i>ECCV 2026 workshop on Computer Vision for Ecology</i> , Malmö
2026	Keynote <i>Open-Earth-Monitor Global Workshop 2026</i> , Barcelona
2026	Keynote <i>ELLIS Winter School 2026: AI for Earth System, Hazards & Climate Extremes</i> , Athens
	Fusing Remote Sensing Data Across Sensors and Time
2025	Seminar talk <i>Wageningen University & Research</i> , Wageningen
	Learning From Global Earth Observation Data
2025	Keynote <i>OpenGeoHub summer school “Data Science for Earth observation”</i> , Wageningen
2025	Keynote <i>QIM Workshop on “Challenges in analysis of large image data”</i> , SCIA2025, Reykjavík
2024	Invited talk <i>Winter school “BigAI for BigGIS”</i> , Indian Institute of Remote Sensing, Webinar
2024	Invited talk <i>Workshop LifeCLEF2024</i> , Grenoble
2024	Seminar talk <i>Allen Institute for AI (AI2)</i> , Seattle
2024	Seminar talk <i>Danish Hydraulic Institute (DHI)</i> , Hørsholm
	MMEarth: Multi-Modal Pretext Tasks For Geospatial Representation Learning
2025	Invited talk “ <i>From Pixels to Products: CV in Danish Research & Industry</i> ”, D3A, Nyborg
2025	Invited talk <i>Nordic Workshop on AI for Climate Change</i> , Gothenburg
2025	Talk <i>ESA-NASA International Workshop on AI Foundation Model for EO</i> , Frascati
	Global Vegetation Monitoring With Probabilistic Deep Learning
2023	Invited talk <i>RISE Learning Machines Seminars</i> , Lund

2023	Invited talk <i>AI for Good</i> , Webinar
2023	Keynote <i>PhD course on “Deep learning in forest remote sensing”</i> , Ås
2022	Invited talk <i>ML4RS: Machine learning for remote sensing reading group</i> , Webinar
	Forest Canopy Height Mapping at Global Scale by Fusing Sentinel-2 and Gedi
2022	Talk <i>ESA Living planet symposium</i> , Bonn
	Global Vegetation Height Mapping With Deep Ensembles
2022	Seminar talk <i>Alan Turing Institute</i> , Webinar
	High Carbon Stock Mapping at Large Scale
2022	Lightning talk <i>Google Geo for Good</i> , Webinar
	Towards Global High-Resolution Biomass Maps: Machine Vision at Large Scale
2021	Seminar talk <i>ESA Φ-Lab</i> , Webinar
2021	Invited talk <i>High carbon stock approach (HCSA) Largescale Working Group</i> , Webinar
	Machine Learning for Environmental Sciences
2019	Invited talk <i>WSIS Forum Workshop “Analysis of aerial data for Aid & Development”</i> , Geneva

Panel discussions

2026	<i>Career panel at the ELLIS Winter School 2026</i> , Athens
2025	<i>Nordic–Alan Turing Institute AI Research Workshop</i> , Oslo
2025	<i>OpenGeoHub Summer School “GeoAI – Dos and Don’ts”</i> , Wageningen
2025	<i>D3A Conference Session “From Pixels to Products”</i> , Nyborg
2024	<i>ICLR 2024 workshop ML for Remote Sensing, “Beyond Benchmarks: ML for the Planet”</i> , Vienna

Awards

2023	Culmann Prize (Outstanding doctoral thesis, ETH Zürich)
2023	CVPR Outstanding Reviewer
2019	U.V. Helava Award (Best Paper, ISPRS Journal)

Selected news media

For a complete list please refer to my [personal website](#).

2022	German national TV ARD quiz show: Wer Weiss Denn Sowas
2022	Swiss national TV SRF news “10 vor 10”: Living Planet Symposium mit Schweizer Beteiligung
2022	NASA: Scientists Show How Forests Measure Up
2022	NVIDIA: Neural Network Generates Global Tree Height Map, Reveals Carbon Stock Potential
2022	ETH Zürich: Neural network can read tree heights from satellite images
2021	SRF1 Swiss national radio live interview: Die Vermessung der Wälder
2021	Mongabay: Chocolate giant funds high resolution carbon map to protect forests
2020	ETH Zürich Industry Relations: A global tool against deforestation

Funding awarded (Total: ≈\$355K)

2026	<i>Horizon Europe</i> - EarthGenerator a foundation model for Earth system modelling, (with Christian Igel), 282K Euro (Total project 6M Euro)
2025	<i>Danish Data Science Academy</i> - Small event grant for the AICC EurIPS workshop, \$7600.
2025	<i>Pioneer Centre for AI</i> - P1 program on AI for Climate & Conservation, \$7700
2025	<i>Google DeepMind</i> - Sponsorship for the FGVC12 workshop at CVPR (with Serge Belongie and Christine Kaeser-Chen), \$5000
2025	<i>Pioneer Centre for AI</i> - Research assistant 4 months (part-time), \$10,500 (in-kind) Project: Neural Implicit Fields for Multi-Image Super-Resolution
2024	<i>Pioneer Centre for AI</i> - Part-time research assistant 2 months, \$1,250 (in-kind) Project: MMEarth-Bench: Global Environmental Tasks for Multimodal Geospatial Models
2024	<i>University of Copenhagen</i> - SSL4EO-2024 PhD summer course (with Stefan Oehmcke, Ankit Kariryaa, Christian Igel), \$6800

2023	<i>Pioneer Centre for AI</i> - Research assistant 6 Months, \$39,500 (in-kind) Project: MMEarth: Exploring Multi-Modal Pretext Tasks For Geospatial Representation Learning
2023	<i>University of Copenhagen and University of Edinburgh (UoE)</i> - Strategic Partnership seed-fund (with Serge Belongie, Oisin Mac Aodha, Stella Frank), \$17,700 Financed the Visipedia workshop 2024 and my 3-weeks research visit at UoE in April 2024.
2021	<i>Amazon</i> - Cloud computing credits (with Peter Gehler, Jan Dirk Wegner) \$20,000 Project: A high-resolution canopy height model of the Earth
2018	<i>ETH Global</i> - Travel grant to represent ETH at the Clinton Global Initiative University meeting in Chicago, (with Andrés C Rodríguez), \$2,000
2016	<i>Degen Stiftung ETH Zürich</i> - Travel award for summer research visit at Caltech (USA), \$1,150

Supervisor functions

Co-supervised 12 master’s and bachelor’s theses. Co-advised 8 PhD students.

Teaching experience

09/2025	Lecturer , <i>OpenGeoHub Summer School “Data Science for EO”</i> , Wageningen
07/2024	Lecturer , <i>SSL4EO-2024 PhD Summer Course</i> , University of Copenhagen
02/2018–05/2022	Teaching Assistant , <i>Photogrammetry and Remote Sensing</i> , ETH Zürich
03/2013–09/2017	Teaching Assistant , ETH Zürich
08/2010–08/2017	Volleyball Instructor , <i>J+S School Sport</i> , Switzerland

Tutorials on “Deep Learning for Geospatial Data Analysis”

09/2019	Tutorial, <i>ECML/PKDD Summer School (EPSS19)</i> , Würzburg
06/2019	Lecture & Tutorial, <i>ISPRS Geospatial Week</i> , Enschede
01/2019	Tutorial, <i>Workshop on ML for Environmental and Geosciences (MLEG2019)</i> , Zürich
01/2018	Tutorial, <i>ISPRS TC II Symposium</i> , Riva

Selected mentorships

2024-	PhD student (University of Agder & The Norwegian Mapping Authority) Co-supervising Sander Riisøen Jyhne during his one-year research visit at the University of Copenhagen. Together we are working on implicit neural representations for multi-image super-resolution. Conference submission under review.
2024-	PhD student (Harvard & University of Copenhagen) Co-supervising Lucia Gordon (with Andrew Davies, Harvard) since her summer visit at the Pioneer Centre for AI in 2024. As a follow-up of the MMEarth project, we are working on the MMEarth-Bench project. Conference submission is under review.
2024-	PhD student (McGill University & Mila Quebec AI Institute) Co-advising Yuyan Chen (with David Rolnick, McGill & Sara Beery, MIT) working on benchmarking open-set recognition and category discovery for insect monitoring. Publication at NeurIPS 2025, Spotlight.
2023-	PhD student (University of Copenhagen) Co-advising Hui Zhang at together with Christian Igel. Hui is working on an extension of my PhD research towards mapping the full vertical vegetation structure. I have listed her for a named two-year postdoc position in the RECLAIM proposal (under review at NordForsk, starting in early 2026).
2024-	Research Assistant (Pioneer Centre for AI) Supervised Vishal Nedungadi as a research assistant after his master, working on the MMEarth project for 6 months at the Pioneer Centre for AI, which was published at ECCV 2024. I mentored Vishal during the application for PhD positions. Co-advising Vishal in his PhD research at Wageningen University & Research.
2023-2025	SURF undergrad (Caltech) Supervised Avirath Sundaresan during a summer research visit as a SURF fellow. Mentored Avirath during weekly meetings together with Sara Beery (MIT) working on animal re-identification.
2023-2024	Bachelor students (University of Copenhagen) Co-supervised Philip Enevoldsen and Christian Gundersen during their Bachelor thesis. Mentored them beyond their thesis to present at the ICCV 2023 workshop OOD-CV and the NLDL 2025 conference leading to a publication in PMLR proceedings, supported by a DDSA travel grant.

Programming & Software

Python: PyTorch, Rasterio, GeoPandas

Other bash, Slurm, GDAL, QGIS

Languages

English (fluent), German (native), Swiss-german

References

Serge Belongie

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Director, Pioneer Centre for AI

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Sara Beery

Massachusetts Institute of Technology

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Co-lead, Global Center on AI and Biodiversity Change

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Oisin Mac Aodha

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Reader (Associate Professor) in Machine Learning

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Ralph Dubayah

University of Maryland

Professor of Geographical Sciences

NASA GEDI Principal Investigator

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